

Consult #5

14/7/2026

Machine Learning Model Dataset

Target

no	Target	task
1	DO	Regression
2	BOD	Regression
3	Turbidity	Regression
4	SS	Regression
5	NH3-N	Classification
6	TCB	Regression
7	FCB	Regression

Dataset Filtered Condition

All data in model dataset filtered by these condtioins from Sentinel-2

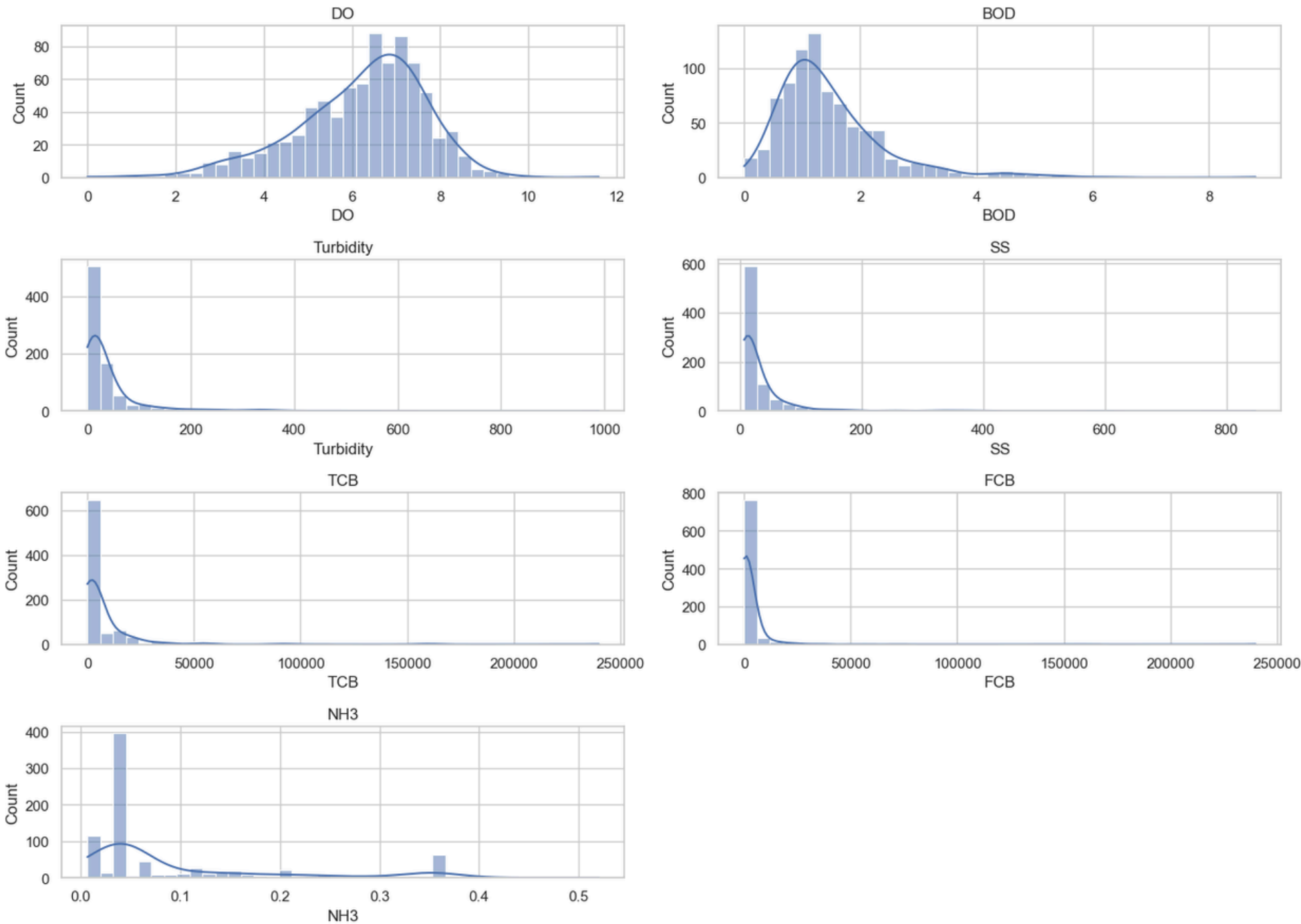
- window 3d or fallback 7d
- Water fraction >0%
- Valid water pixel in buffer > 0
- Local cloud <= 70%
- B3_median not NaN

NH3-N class

Low: <=0.05

Medium: >0.05 - <=0.12

High: >0.12



Machine Learning Model Dataset

Validation Split

1.Temporal Split
Train: 2562-2566
Test: 2567-2568

Train: 2562-2565
Validation: 2566
Test: 2567-2568

2.Station Wise split

Validation

1.Regression

- MAE
- RMSE
- R²
- Pearson r
- Spearman r
- Bias

2.Classification

- accuracy
- Macro F1
- Confusion matrix

Feature Set

- 1.Sentinel-2 Features only
- 2.Sentinel-2 Features + Context Features
- 3.Context Features only (experiment)

Experiment	Split	Feature set	Models	Targets	จำนวนผล
AutoGluon main	Temporal	S2 + Context, S2 only, Context only	WeightedEnsemble L2/L3	6 regression + NH3	21
Manual temporal	Temporal	S2 only, Context only, S2 + Context	Ridge, ElasticNet, RF, ExtraTrees,	6 regression + NH3	186
Manual station-temporal	Station + Temporal	S2 only, Context only, S2 + Context	same manual models	6 regression + NH3	558

Best Model overall

Target	Best experiment	Feature set	Model	Transform	MAE	RMSE	R²	Spearman r	Bias
DO	Manual temporal	S2 + Context	RandomForest	raw	0.732	0.89	0.42	0.593	-0.237
BOD	AutoGluon temporal	S2 only	WeightedEnsemble_L3	raw	0.507	0.726	0.29	0.517	-0.07
Turbidity	AutoGluon temporal	S2 + Context	WeightedEnsemble_L2	log1p	22.678	53.618	0.274	0.709	-12.853
SS	Manual temporal	S2 + Context	Stacking_RidgeMeta	log1p	17.037	52.404	0.262	0.695	-9.759
TCB	Manual temporal	S2 + Context	CatBoost	log1p	4,045	8,476	0.059	0.656	-2,585
FCB	Manual temporal	S2 + Context	ExtraTrees	log1p	1,383	3,156	0.004	0.568	-1,065

Target	Model-scale RMSE	Model-scale R²	Model-scale Spearman
Turbidity	0.766	0.484	0.709
SS	0.619	0.51	0.695
TCB	1.442	0.406	0.656
FCB	1.643	0.359	0.568

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log-scale metric

Best Model non-autogluon

Target	Feature set	Best model	Transform	n_train	n_test	MAE	RMSE	R²	Pearson r	Spearman r	Bias
DO	S2 + Context	RandomForest	raw	276	114	0.732	0.89	0.42	0.681	0.593	-0.237
BOD	S2 + Context	Stacking_RidgeMeta	raw	276	114	0.547	0.77	0.2	0.474	0.438	-0.098
Turbidity	S2 + Context	Stacking_RidgeMeta	log1p	276	114	21.792	54.053	0.262	0.68	0.697	-16.662
SS	S2 + Context	Stacking_RidgeMeta	log1p	276	114	17.037	52.404	0.262	0.673	0.695	-9.759
TCB	S2 + Context	CatBoost	log1p	276	114	4,045	8,476	0.059	0.386	0.656	-2,585
FCB	S2 + Context	ExtraTrees	log1p	276	114	1,383	3,156	0.004	0.411	0.568	-1,065

Target	Best model	Model-scale RMSE	Model-scale R²	Model-scale Spearman r
Turbidity	Stacking_RidgeMeta	0.77	0.48	0.697
SS	Stacking_RidgeMeta	0.619	0.51	0.695
TCB	CatBoost	1.442	0.406	0.656
FCB	ExtraTrees	1.643	0.359	0.568

log-scale metric

Station-Temporal Robustness

Target	Best feature set	Best model	Metric used	RMSE / log-RMSE	R ²	Spearman r
DO	S2 only	Ridge	original RMSE	1.235	-0.312	0.235
BOD	Context only	Ridge	original RMSE	0.763	0.075	0.346
Turbidity	S2 only	ElasticNet	log RMSE	0.812	0.37	0.63
SS	S2 only	Ridge	log RMSE	0.706	0.278	0.645
TCB	S2 + Context	CatBoost	log RMSE	1.748	0.043	0.388
FCB	S2 only	Stacking_RidgeMeta	log RMSE	2.014	-0.052	0.164

NH-3

Setup	Feature set	Model	Accuracy	Balanced accuracy	Macro F1
AutoGluon NH3	Context only	WeightedEnsemble_L2	0.307	0.38	0.251
Manual temporal NH3 3-class	Context only	Logistic	0.447	0.424	0.425
Manual temporal NH3 binary	S2 + Context	ExtraTrees	0.614	0.683	0.613
Station-temporal NH3 3-class	S2 + Context	Stacking_Logistic Meta	0.447	0.415	0.412
Station-temporal NH3 binary	S2 + Context	ExtraTrees	0.632	0.694	0.627